

- Set : It is a collection of well defined
" { } " objs, elements, things

eg: $S = \{2, 4, 6, 8\}$

Types $\rightarrow$ singleton set : $S = \{5\}$

→ Empty / Null set : $S = \{ \}$ or ϕ

→ finite set : $S = \{2, 4, 6, 8\}$

→ Infinite set : set of Natural Numbers
 $\{1, 2, 3, 4, 5, \dots, \infty\}$

→ Subset: $S = \{1, 2, 3\}$, $A = \{1, 2, 3, 4, 5\}$
 $\therefore S \subseteq A$

→ Powerset: $A = \{1, 2\}$

$$P(A) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}$$

$n \rightarrow$ no. of elements (A)

$$2^n \rightarrow \text{no. of elements (P(A))}$$

→ Universal set: $A = \{1, 2\}$ $B = \{4, 5\}$

$$C \{ 6, 7 \}$$
$$U = \{1, 2, 4, 5, 6, 7\}$$

→ Disjoint set : $A = \{1, 2\}$ $B = \{4, 5\}$

$$A \cap B = \emptyset$$

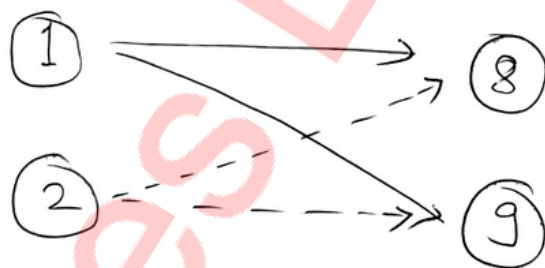
• Basic Operations of set.

① Union ($\cup$) ② Intersection ($\cap$) ③ Difference ($-$)

$$\begin{array}{ccc} A = \{1, 2, 3, 4\} & & B = \{3, 4, 5, 6\} \\ \downarrow & & \downarrow \\ A \cup B & & A \cap B \\ = \{1, 2, 3, 4, 5, 6\} & & = \{3, 4\} \\ & & \downarrow \\ & & A - B \\ & & = \{1, 2\} \end{array}$$

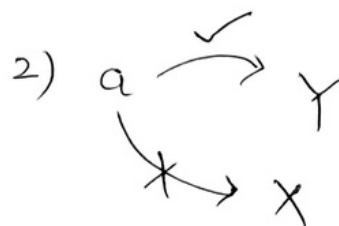
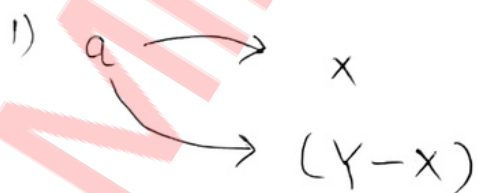
• Cartesian product of 2 sets ($\times$)

$$X = \{1, 2\} \quad Y = \{8, 9\}$$



$$X \times Y = \{(1, 8), (1, 9), (2, 8), (2, 9)\}$$

Eg: $X \cap (Y - X) = \emptyset$



$$\therefore X \cap (Y - X) = \emptyset$$

• Relation: subset of $X \times Y$

$$X = \{1, 2\}$$

$$Y = \{8, 9\}$$

$$X \times Y = \{(1, 8), (1, 9), (2, 8), (2, 9)\}$$

$$\rightarrow \{(1, 8)\}$$

$$\rightarrow \{(1, 9)\}$$

$$\rightarrow \{(1, 8), (1, 9)\}$$

$$\rightarrow \{(1, 8), (2, 8), (2, 9)\}$$

$$X = 'm'$$

$$Y = 'n'$$

$$\text{Total no. of Relations} = 2^{m+n}$$



Types of Relations

- Empty
- Universal
- Identity
- Inverse
- Reflexive
- Symmetric
- Transitive
- Equivalence
- Antisymmetric

① Empty : No relation exist betⁿ any item of a set

eg: $A = \{1, 2\}$ $B = \{8, 9\}$

$$R \rightarrow (a, b) \in R \text{ if \& only if } a = b$$

$$R = \{\}$$

② Universal (Full) : $R = A \times A$

If each element of A is related to every (Have all elements)

eg: "height of students greater than 5cm"

③ Identity : $A = \{1, 2, 3\}$

$$I = \{(1, 1) (2, 2) (3, 3)\}$$

④ Inverse : R^{-1}

$$R = \{(1, 2) (2, 5) (6, 4)\}$$

$$R^{-1} = \{(2, 1) (5, 2) (4, 6)\}$$

⑤ Reflexive : $A = \{1, 2, 3\}$ "(a, a)"

$$A \times A = \{ \text{9 pairs} \}$$

$$R = \{(1, 1), (2, 2), (3, 3)\}, R = \{(1, 1)\}$$

$$R = \{(1,1) (2,2) (3,3)\}$$

$$A = \{1, 2, 3\}$$

$$A \times A = \left\{ \begin{array}{ccc} (1,1) & \text{---} & \text{---} \\ \text{---} & (2,2) & \text{---} \\ \text{---} & \text{---} & (3,3) \end{array} \right\} N^2$$

Diagonal items = 3 i.e. N

$\therefore$ Total Reflexive Relations possible = $2^{N^2 - N}$

$\rightarrow$ Min : N

$\rightarrow$ Max : N^2

⑥ Symmetric : $A = \{1, 2\}$ $B = \{1, 2, 4\}$

$$R = \{(1,2) (2,1)\}$$

$$R = \{(1,2) (2,1) (2,2)\}$$

$$R = \{\}$$

⑦ Antisymmetric : $(a,b) \in R$ $(b,a) \in R$
 $\text{Pf } a = b$

$$A = \{1, 2, 3\}$$

$$R = \{(1,2) (2,1)\} \times R = \{\} \checkmark$$

$$R = \{(1,2) (1,1)\} \checkmark$$

⑧ Transitive : $A \rightarrow B, B \rightarrow C, A \rightarrow C$
 $\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $(A, B) \quad (B, C) \quad (A, C)$

$$R_1 = \{ \}$$

$$R_2 = \{ (1, 1) \}$$

$$R_3 = \{ (1, 2) (2, 1) (1, 1) \}$$

$$R_4 = \{ (5, 6) (6, 7), (5, 7) \}$$

⑨ Equivalence Relation : $(R) + (S) + (T)$

$$X = \{ a, b, c \}$$

$$R_1 = \{ \}$$

$$R_2 = \{ (a, a) (b, b) (c, c) \}$$

$$R_3 = \{ (a, a) (b, b) (c, c) (b, a) (a, b) \}$$

$$R_4 = \{ (a, a) (b, b) (c, c) (a, b) \}$$

→ POSET : Partial ordering Relation

Partially ordered Set $\nearrow$ (R + AntiSy + T)

Relation

→ $[X, R]$

$$X = \{ a, b, c \}$$

$$R_1 = \{ (a, a) (b, b) (c, c) \}$$

$$R_2 = \{ (a, a) (b, b) (c, c) (a, b) (b, c) (a, c) \}$$

$$R_3 = \{ \}$$

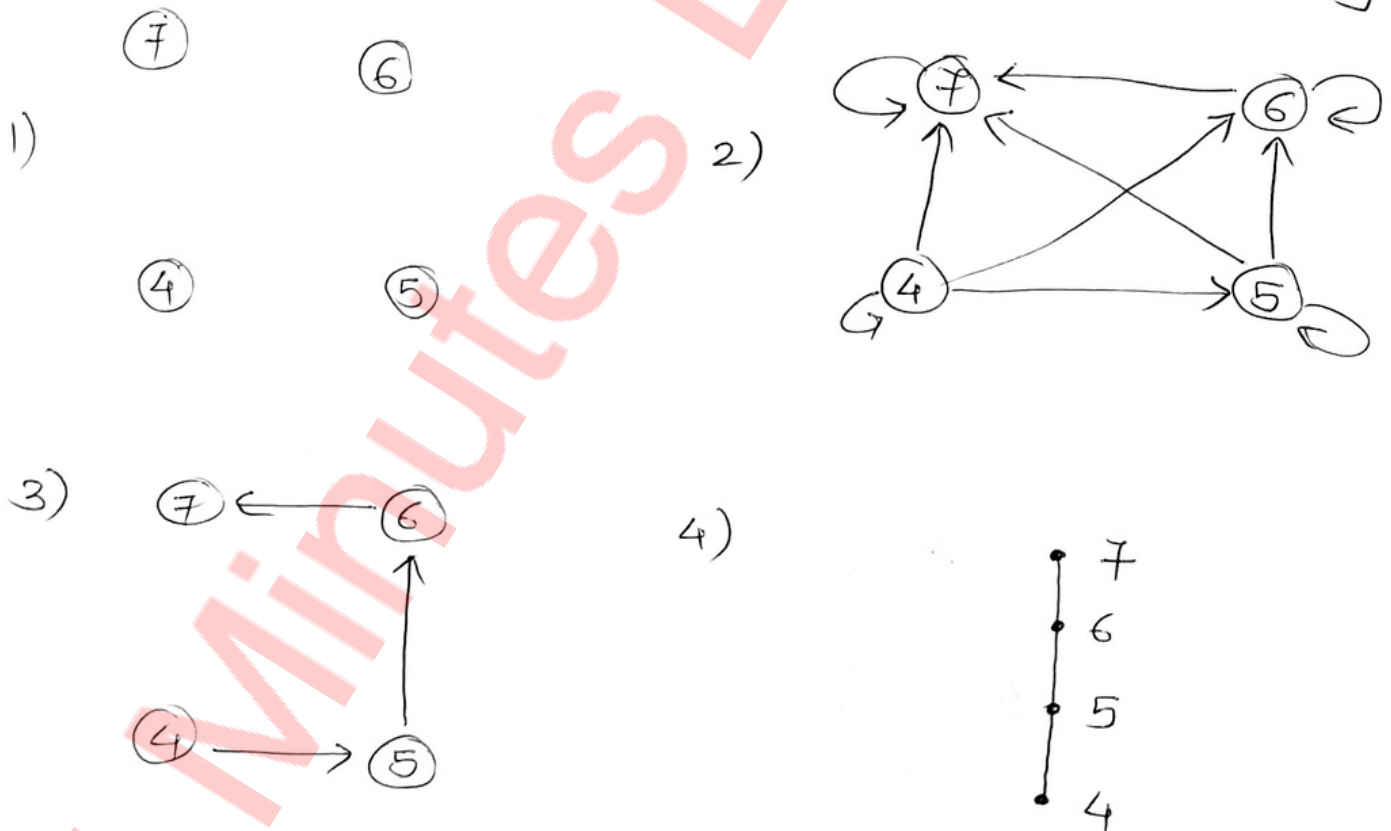
• Hasse Diagram

Q: Convert POR into hasse diagram

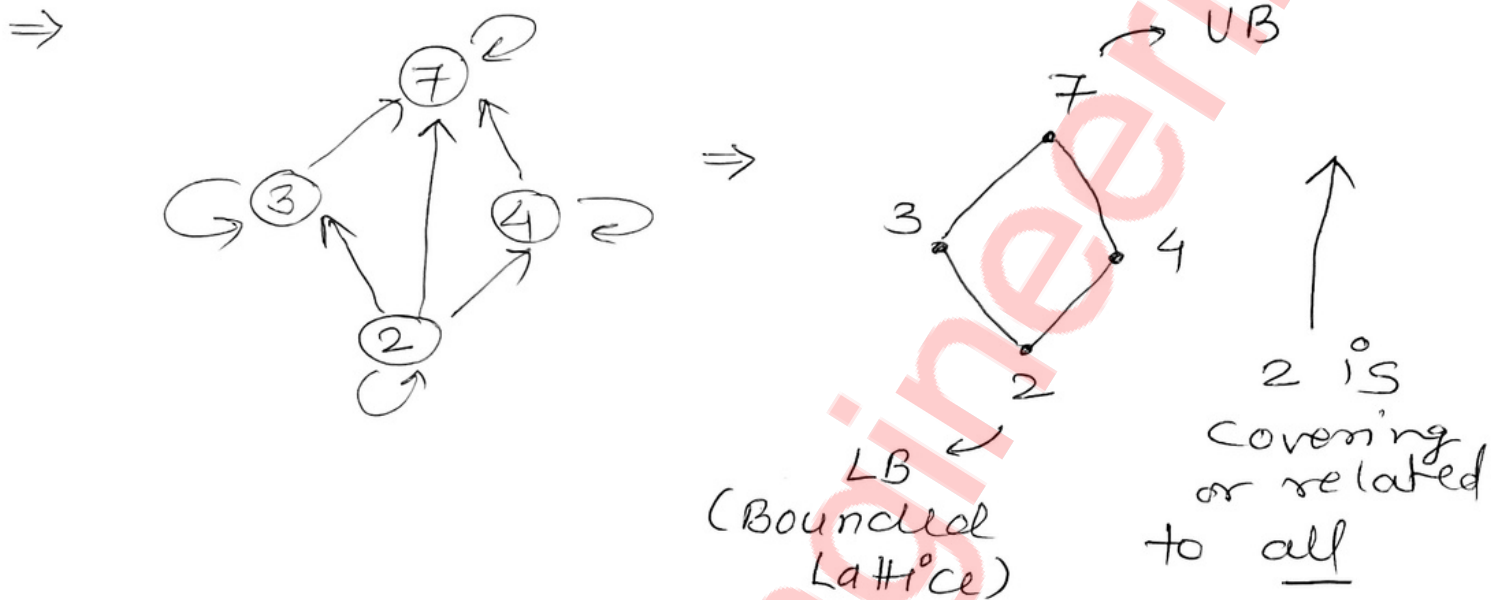
- 1) $\rightarrow$ Plot a vertex for every element of set
- 2) $\rightarrow$ Draw a edge betⁿ x & y if $(x, y) \in R$
- 3) $\rightarrow$ No need of Reflexive & Transitive edges
Remove them.
- 4) $\rightarrow$ No directions needed (edges)
(upward orientation)

eg: Set $A = \{4, 5, 6, 7\}$

$$R = \{(4, 5) (4, 6) (4, 7) (5, 6) (5, 7) \\ (6, 7) (4, 4) (5, 5) (6, 6) (7, 7)\}$$



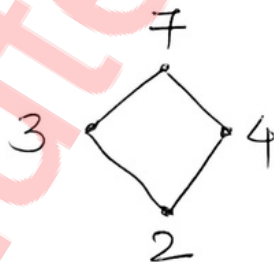
eg: $R = \{(2,2), (2,3), (2,4), (2,7), (3,3), (3,7), (4,4), (4,7), (7,7)\}$
 $A = \{2, 3, 4, 7\}$



Lattice : It is hasse diagram with a condition

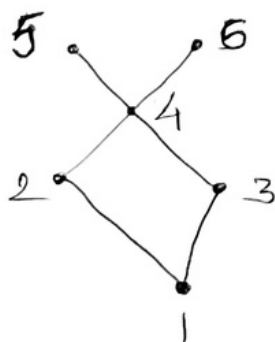
$\Rightarrow$ (Join & Meet for each pair of element)

eg:



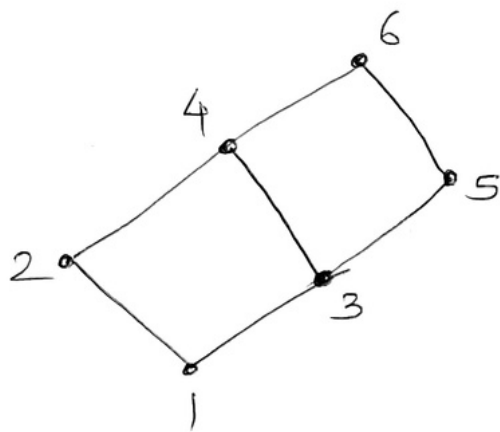
Join
 $\downarrow$
 $3 \vee 4 = 7$
 $3 \wedge 4 = 2$
 $\uparrow$
 meet

eg:



$2 \vee 3 = 4$
 $2 \wedge 3 = 1$
 but $5 \vee 6 = \underline{\quad}$
 $5 \wedge 6 = 4$

eg:



$$4 \vee 5 = 6$$

$$4 \wedge 5 = 3$$

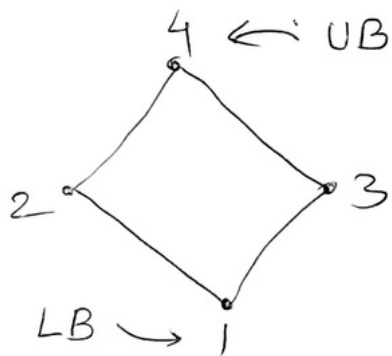
$$3 \vee 5 = 5$$

$$3 \wedge 5 = 3$$

$$2 \vee 3 = 4$$

$$2 \wedge 3 = 1$$

eg:



Finding complement.

$$2 \vee 3 = 4 \leftarrow \text{UB}$$

$$2 \wedge 3 = 1 \leftarrow \text{LB}$$

So 2 & 3 are complement of each other.

$$\begin{aligned} e \vee e^c &= \text{UB} \\ e \wedge e^c &= \text{LB} \end{aligned}$$

Distributive $\rightarrow$ (0 or 1) at most 1

Complement $\rightarrow$ (1 or more) at least

• Totally ordered set

↳ POSET + All elements in set are comparable with each other.

' $\leq$ '

eg: $X = \{a, b, c\}$

Relation $(R) \rightarrow a \leq b, a \leq c, b \leq c$

eg: Divide($/$) in set of whole numbers.

$[(2/2)] [(10/5) (5/10)]$

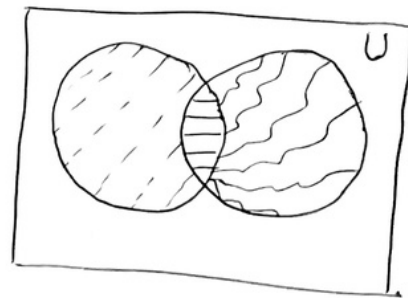
$[(8/4) (4/2) \rightarrow (8/2)]$

$[4/3]$

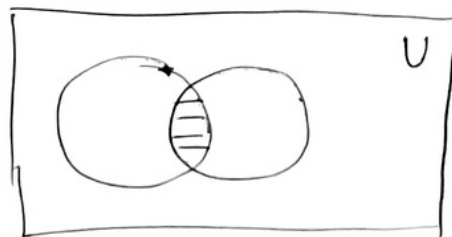
• Venn Diagram

$U: \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A: \{1, 2, 3\}$ $B: \{3, 4, 5\}$

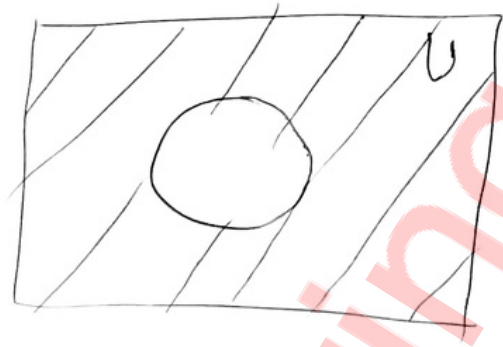
$A \cup B: \{1, 2, 3, 4, 5\}$



$A \cap B: \{3\}$



$$A^C = \{4, 5, 6, 7, 8, 9, 10\}$$

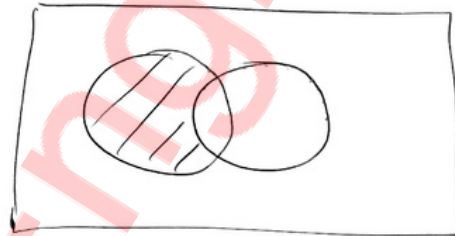


• Symmetric difference $\oplus$ (not for both)

$$A \oplus B = \{1, 2, 4, 5\}$$

• Difference $(-)$

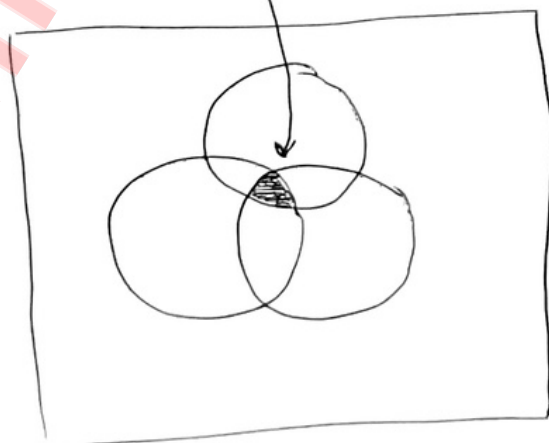
$$A - B = \{1, 2\}$$



$$(A \cup B) - B$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \{1, 2, 3, 4, 5\} & - & \{3, 4, 5\} \\ \downarrow & & \\ \{1, 2\} & & \end{array}$$

$$\rightarrow (A \cap B \cap C)$$



→ Multiset : { Multiple copies of elements in a set }

Eg: $[1, 1, 2, 2, 3, 3]$

Multiplicity $\left\{ \begin{array}{l} 1 \rightarrow 2 \\ 2 \rightarrow 2 \\ 3 \rightarrow 2 \end{array} \right\}$

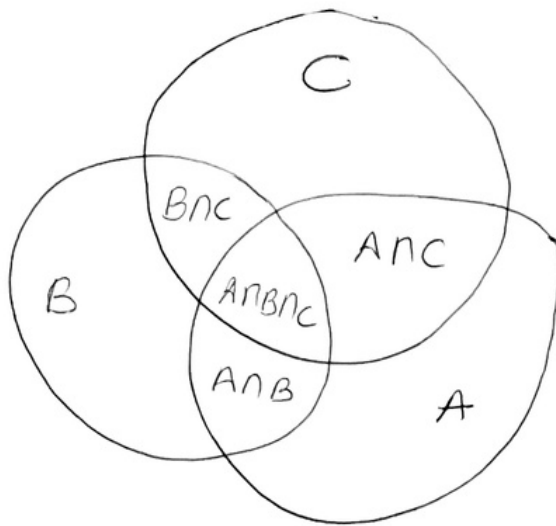
$A = [1, 1, 2, 2, 3, 4]$ $B = [1, 1, 2, 2, 6, 6, 6]$

$A \cup B$ $\rightarrow [1, 1, 2, 2, 3, 4, 6, 6]$

$A \cap B$ $\rightarrow [1, 1, 2, 2]$

$A - B$ $\rightarrow [3, 4]$

→ Inclusion and Exclusion principle



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

eg: $n(S_1) \rightarrow 49$ $n(S_1 \cap S_2) \rightarrow 9$
 $n(S_2) \rightarrow 37$ $n(S_2 \cap S_3) \rightarrow 5$ $n(S_1 \cap S_2 \cap S_3) = 3$
 $n(S_3) \rightarrow 21$ $n(S_1 \cap S_3) \rightarrow 4$

To find:

$$\begin{aligned} n(S_1 \cup S_2 \cup S_3) &= n(S_1) + n(S_2) + n(S_3) \\ &\quad - n(S_1 \cap S_2) - n(S_1 \cap S_3) - n(S_2 \cap S_3) \\ &\quad + n(S_1 \cap S_2 \cap S_3) \\ &= 49 + 37 + 21 - 9 - 4 - 5 + 3 \\ &= \underline{92} \end{aligned}$$

$$\begin{aligned} n(S_1 \cup S_2 \cup S_3)^c &= U - n(S_1 \cup S_2 \cup S_3) \\ &= 100 - 92 = 8 \end{aligned}$$

eg: $U = 120$

$$n(A \cup B \cup C) = 100$$

$$n(A) = 65$$

$$n(B) = 45$$

$$n(C) = 42$$

$$n(A \cap B) = 20$$

$$n(A \cap B \cap C) = ?$$

$$n(A \cap C) = 25$$

$$n(B \cap C) = 15$$

$$\Rightarrow 100 = 65 + 45 + 42 - 20 - 25 - 15 + x$$

$$= 92 + x$$

$$x = 100 - 92$$

$$\boxed{x = 8} \therefore n(A \cap B \cap C) = 8$$

• Mathematical Induction

- ① $n=1$
- ② $n=k$ (k is any +ve integer)
- ③ $n=k+1$

eg: $1 + 3 + 5 + \dots + (2n-1) = n^2$

① L.H.S ($1 + 3 + 5 + \dots + (2n-1)$)

$$\downarrow$$

$$2(1)-1 = 1$$

RHS (n^2)

$$\hookrightarrow (1)^2 = 1$$

$$\therefore \text{L.H.S} = \text{R.H.S} \quad [\text{True for } n=1]$$

② Assume result is true for $n=k$

$$1 + 3 + 5 + \dots + (2k-1) = k^2$$

③ check it for $n=k+1$

$$1 + 3 + 5 + \dots + (2(k+1)-1) = (k+1)^2$$

↓

$$\underbrace{1 + 3 + 5 + \dots + (2k-1)}_{\downarrow} + (2(k+1)-1) = (k+1)^2$$

↓

$$k^2 + (2(k+1)-1) = (k+1)^2$$

$$k^2 + 2k + 2 - 1 = (k+1)^2$$

$$k^2 + 2k + 1 = (k+1)^2$$

$$(k+1)^2 = (k+1)^2$$

$$L.H.S = R.H.S \quad [\text{True for } n=k+1]$$

eg: $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

① L.H.S; $n=1$

R.H.S; $\frac{1(1+1)}{2} = \frac{2}{2} = 1 \quad [\text{True for } n=1]$

② Assume result is true for $n=k$

$$1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2}$$

③ $n = k + 1$

$$1 + 2 + 3 + \dots + (k+1) = \frac{(k+1)(k+1+1)}{2}$$

↓

$$\underbrace{1 + 2 + 3 + \dots + k + (k+1)}_{\downarrow} = \frac{(k+1)(k+2)}{2}$$

$$\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$$

$$\frac{(k+1)(k+2)}{2} = \frac{(k+1)(k+2)}{2}$$

$$L.H.S = R.H.S$$

$$[\text{True for } n = k + 1]$$

• Propositional Logic



Basic unit of Logic

It's a statement/sentence that can be T or F but not both.

$$2 + 3 = 5 \checkmark$$

"Hello & welcome Dosto" ✗

Connectives (operators)

→ NOT/ Negation ($\sim$ $\neg$)

↪ given $\rightarrow$ Result ($\neg$)

T	F
F	T

→ Conjunction ($\wedge$) (and)

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

→ Disjunction ($\vee$) (or)

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

→ Implication ($\rightarrow$)

If A then B ($A \rightarrow B$)

A	B	$A \rightarrow B$
T	T	T
T	F	F $\leftarrow$
F	T	T
F	F	T

→ Exclusive Or: ' $\oplus$ '

A	B	$A \oplus B$
T	T	F
T	F	T
F	T	T
F	F	F

→ Double Implication: ' $\leftrightarrow$ ' Same (T)

A	B	$A \leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

⇒ Converse Inverse Contrapositive

"Given $A \rightarrow B$ "

$B \rightarrow A$ $\sim A \rightarrow \sim B$ $\sim B \rightarrow \sim A$

A	B	$\sim A$	$\sim B$	$A \rightarrow B$	$B \rightarrow A$	$A \leftrightarrow B$
T	T	F	F	T	T	T
T	F	F	T	F	T	F
F	T	T	F	T	F	F
F	F	T	T	T	T	T

$\sim A \rightarrow \sim B$
T
T
F
T

$\sim B \rightarrow \sim A$	$\sim A \vee B$
T	T
F	F
T	T
T	T

$$\therefore A \rightarrow B = \sim B \rightarrow \sim A$$

$$B \rightarrow A = \sim A \rightarrow \sim B$$

$$A \rightarrow B = \sim A \vee B$$

} Imp

Tautology : (All True)

Contradiction: (All False)

Contingency : (both True false)

eg: $(A \wedge B) \wedge \sim (A \vee B)$

A	B	$A \wedge B$	$A \vee B$	$\sim (A \vee B)$	'R'
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

Contradiction.

eg:- $[(A \rightarrow B) \wedge A] \rightarrow B$

A	B	$A \rightarrow B$	$(A \rightarrow B) \wedge A$	$[(A \rightarrow B) \wedge A] \rightarrow B$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Tautology.

eg:- $(A \vee B) \wedge (\neg A)$

A	B	$\neg A$	$A \vee B$	$(A \vee B) \wedge (\neg A)$
T	T	F	T	F
T	F	F	T	F
F	T	T	T	T
F	F	T	F	F

Contingency.

Propositional Equivalences.

- ① Same truth values (Truth table match)
- ② $A \leftrightarrow B$ is tautology.

I) $\neg(A \vee B)$

II) $(\neg A) \wedge (\neg B)$

A	B	$\neg A$	$\neg B$	$\neg(A \vee B)$	$(\neg A) \wedge (\neg B)$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

$$[\neg(A \vee B)] \leftrightarrow [(\neg A) \wedge (\neg B)]$$

$$\begin{matrix} T \\ T \\ T \\ T \end{matrix} \} \text{Tautology}$$

◦ Laws.

↳ Idempotence

$$P \vee P = P$$

$$P \wedge P = P$$

◦ Commutative

$$P \vee Q = Q \vee P$$

$$P \wedge Q = Q \wedge P$$

◦ Associative

$$P \vee (Q \vee R) = (P \vee Q) \vee R$$

$$P \wedge (Q \wedge R) = (P \wedge Q) \wedge R$$

◦ Distributive

$$P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$$

$$P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$$

◦ Double negation $\sim(\sim P) = P$

◦ De Morgan

$$\sim(P \wedge Q) = \sim P \vee \sim Q$$

$$\sim(P \vee Q) = \sim P \wedge \sim Q$$

Identity

$$P \vee T = T$$

$$P \vee F = P$$

$$P \wedge T = P$$

$$P \wedge F = F$$

P	$\sim P$	$P \vee \sim P$	$P \wedge \sim P$
T	F	T	F
F	T	T	F
		$\underbrace{\quad\quad\quad}_T$	$\underbrace{\quad\quad\quad}_F$

Normal form



Disjunctive (DNF)

→ Disjunction (OR)
betn
Conjunction (AND)

$$(A \wedge B) \vee (\sim A \wedge B)$$

$$(A \text{ and } B) \text{ OR } (\sim A \text{ and } B)$$

Conjunctive (CNF)

→ Conjunction (AND)
betn
Disjunction (OR)

$$(A \vee B) \wedge (\sim A \vee B)$$

$$(A \text{ OR } B) \text{ and } (\sim A \text{ OR } B)$$

eg:- $(x \rightarrow y) \wedge (y \rightarrow x)$ (find DNF)

$$(\sim x \vee y) \wedge (\sim y \vee x)$$

$$[(\sim x \vee y) \wedge \sim y] \vee [(\sim x \vee y) \wedge x]$$

$$[(\sim x \wedge y) \vee (y \wedge \sim y)] \vee [(\sim x \wedge x) \vee (y \wedge x)]$$

$$[(\sim x \wedge y) \vee F] \vee [F \vee (y \wedge x)]$$

$$\Rightarrow (\sim x \wedge \sim y) \vee (y \wedge x) \rightarrow \underline{\text{DNF}}$$

eg: $A \wedge (A \rightarrow B)$

$$\downarrow$$

$$A \wedge (\sim A \vee B)$$

$$(A \wedge \sim A) \vee (A \wedge B)$$

$$F \vee (A \wedge B)$$

$$\rightarrow (A \wedge B) \rightarrow \text{CNF}$$

$\rightarrow$ Argument Premises Conclusion.
 $(P_1 \wedge P_2 \wedge \dots \wedge P_n)$ "Statements" $(P_1, P_2 \dots P_n)$ $(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow C$

• Modus Ponens

	Given (True)	Find
	$P \rightarrow q$	$P \rightarrow q$
P1: $P \rightarrow q$ (true)	T T	T ✓
P2: P (true)	T F	F
	F T	T
	F F	T
q		

• Modus Tollens

P1: $P \rightarrow q$ (True)
 P2: $\sim q$ (False)

 $\sim P$ (False)

here q & P are false (given)

• Hypothetical Syllogism

P₁: $P \rightarrow q$ (True)

P₂: $P \rightarrow r$ (True)

$P \rightarrow r$ (True)

$(P \rightarrow q) \wedge (P \rightarrow r) \rightarrow (P \rightarrow r)$

• Disjunctive Syllogism

P₁: $P \vee q$

P₂: $\sim P$

q

$(P \vee q) \wedge (\sim P) \rightarrow q$

P	q	$\sim P$	$P \vee q$		
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

✓ { (F, T, T, T) }

eg: A: I will study

B: I will play

P₁: (I will study) or (I will play)

P₂: (I will not study)

(I will play)

P₁: $A \vee B$

P₂: $\sim A$

B

✓ valid Argument

eg:- A :- I study from 5ME

B : I am prepared for exam

C : I will pass my exam.

→ If I study from 5ME, I am prepared for exam. If I am prepared for exam, I will pass my exam. Therefore If I ~~am~~ study from 5ME then I will pass my exam.

P1: $A \rightarrow B$

P2: $B \rightarrow C$

$A \rightarrow C$

valid ✓

• Quantifiers

→ For all ($\forall$) Universal

→ There exists ($\exists$) Existential

$\forall x (x)$

eg: All Engineers
love 5ME

$\exists x f(x)$

Some Engineers
love 5ME

or
atleast one engineer
love 5ME

Negation

$$\forall x f(x) \quad \swarrow \searrow$$
$$\forall x [\sim f(x)]$$

$$\exists x f(x) \quad \swarrow \searrow$$
$$\exists x [\sim f(x)]$$

eg: $P(x) : x$ is student

$Q(x) : x$ is a boy

→ Every child is not a student

$$\forall x [\sim P(x)]$$

→ Every child is a student

$$\forall x [P(x)]$$

→ Every child is girl

$$\forall x [\sim Q(x)]$$

$$\Rightarrow \sim (\forall x f(x)) = \exists x \sim f(x)$$

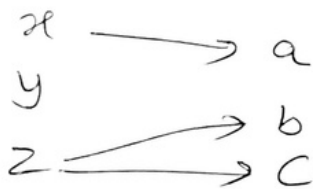
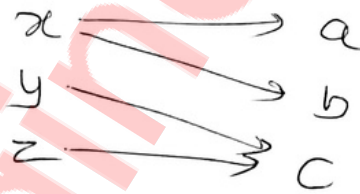
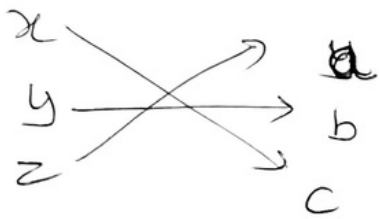
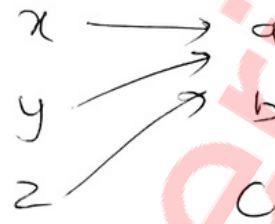
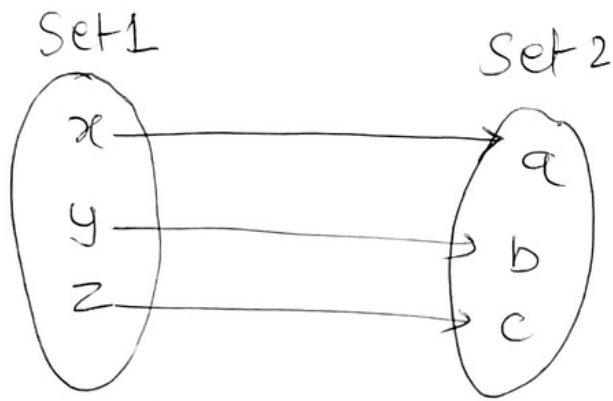
$$\Rightarrow \sim (\exists x f(x)) = \forall x \sim f(x)$$

eg:- not all students are naughty.

$$\sim \exists x [\text{student}(x) \wedge \sim \text{naughty}(x)]$$

$$\forall x [\sim S(x) \vee N(x)] = \forall x [S(x) \rightarrow N(x)]$$

function



$\Rightarrow$

set 1



set 2



$n \times n \times n \dots m \text{ times}$

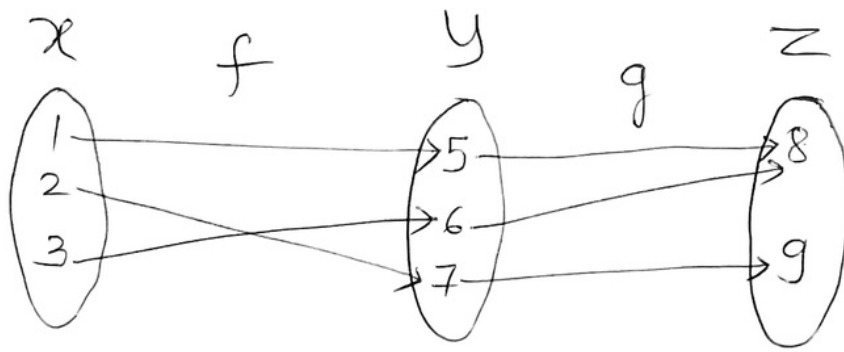
n^m

total f^n possible

• function Composition

$$f(g(x)) = f \circ g(x)$$

$$g(f(x)) = g \circ f(x)$$



$$f = \{(1, 5), (2, 7), (3, 6)\}$$

$$g = \{(5, 8), (6, 8), (7, 9)\}$$

$$f(x) \rightarrow y$$

$$1 \rightarrow 5$$

$$2 \rightarrow 7$$

$$3 \rightarrow 6$$

$$g(y) \rightarrow z$$

$$5 \rightarrow 8$$

$$6 \rightarrow 8$$

$$7 \rightarrow 9$$

$$g \circ f(x) = g(f(x)) = \{(1, 8), (2, 9), (3, 8)\}$$

eg: $f(x) = 2x$

$g(x) = x + 1$

find $f \circ g(x)$ if $x = 1$

$$f \rightarrow g : f \circ g(x) = f(g(x))$$

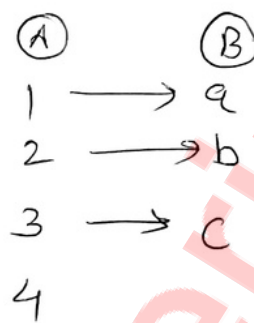
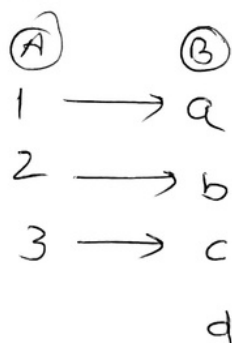
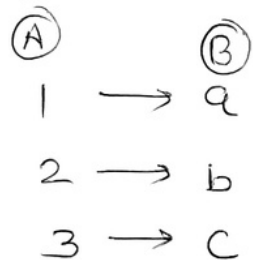
$$= f(x+1)$$

$$= 2(x+1)$$

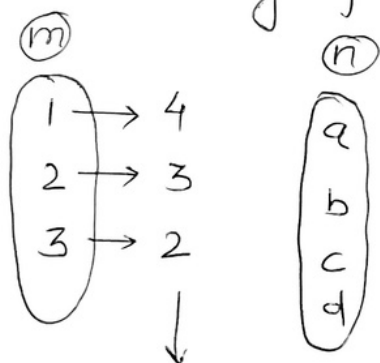
now put $x = 1$

$$\therefore 2(1+1) = 2 \times 2 = 4$$

◦ Injective function (one to one)



Possible no. of fn.

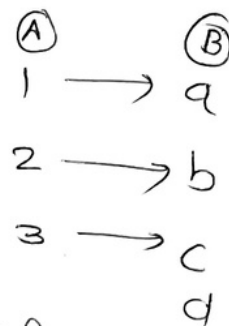
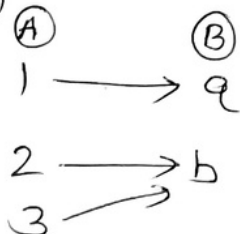
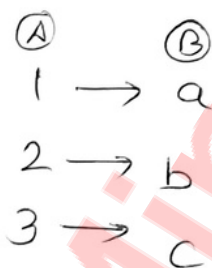


$$4 \times 3 \times 2 = 24$$

$$n(A) \leq n(B)$$

formula = $\frac{n!}{(n-m)!} = \frac{4!}{(4-3)!} = \frac{4!}{1!} = 4 \times 3 \times 2 \times 1 = \underline{24}$

◦ Surjective function (Cover all elements of Range)
(onto)



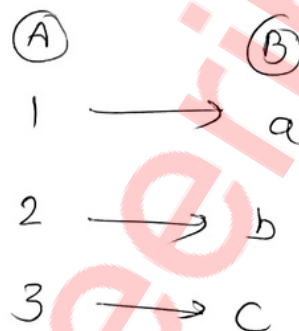
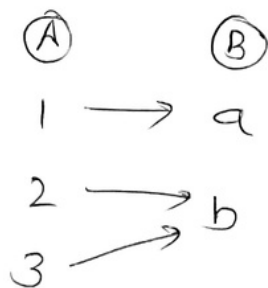
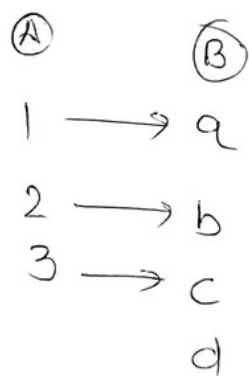
✓
one to one
onto
X

Domain = Range

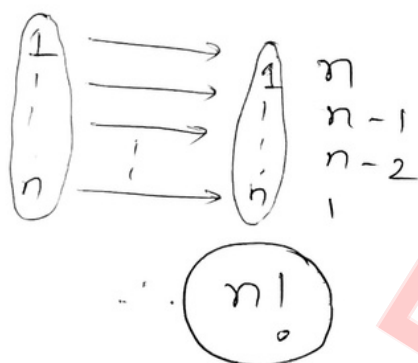
$$n(A) \geq n(B)$$

• Bijjective Function

$$\rightarrow n(A) = n(B)$$



• Possible f^n_s .



• Inverse of function (Reverse).

• $f(x) = y$ then $f^{-1}(y) = x$.

• f^n must be bijective

• Not possible with one to one or onto.

eg: $f(x) = 1 + 2x$

$$f^{-1}(y) = \frac{y-1}{2}$$

Let $x = 1$

$$1 + 2(1) = 3$$



$$\frac{(3-1)}{2} = \frac{2}{2} = 1$$

Combinatorics

Combination

→ order does not matter

eg:- faluda

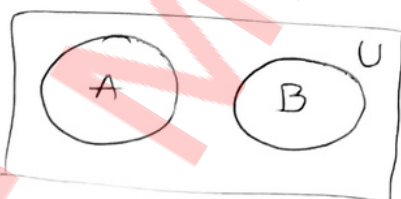


$${}^nC_k = \frac{n!}{k!(n-k)!}$$

eg: Select 5 out of 25

$$\begin{aligned} {}^{25}C_5 &= \frac{25!}{5!(25-5)!} \\ &= \frac{25 \times 24 \times 23 \times 22 \times 21}{5 \times 4 \times 3 \times 2 \times 1} \\ &= 53130 \end{aligned}$$

• Sum Rule
'OR'



$$n(A \cup B) = n(A) + n(B)$$

Permutation

→ order matters

eg: lock-key-password.



$${}_nP_k = \frac{n!}{(n-k)!}$$

eg: set (n items)

↓
want k items
arrangement

(1 2 3 4) n=4

→ — — k=2

$$\begin{aligned} {}_4P_2 &= \frac{4!}{(4-2)!} = \frac{4 \times 3 \times 2 \times 1}{2} \\ &= \underline{12} \end{aligned}$$

Task ①

'm'
choices

Task ②

'n'
choices

↓
 $\frac{m+n}{}$

(2 task cannot be done at same time)

• Product Rule

Task ①

'm'

choices

Task ②

'n'

choices.

→ m × n ←

$$n(A \times B) = n(A) \cdot n(B) \quad (\text{Task ② after Task ①})$$

eg:- 3 digit numbers from digits 1, 2, 3, 4, 5
(Repetition allowed).

⇒

$$\begin{array}{ccc} \frac{5 \times}{H} & \frac{5 \times}{T} & \frac{5 \times}{U} \\ & = 5 \times 5 \times 5 \\ & = 25 \end{array}$$

(without Repetition) (1, 2, 3, 4, 5)

$$\begin{array}{ccc} \frac{\quad}{H} & \frac{\quad}{T} & \frac{\quad}{U} \end{array} \quad \begin{array}{l} \swarrow \\ n = 5 \\ \searrow \\ k = 3 \end{array}$$
$$\therefore {}^5P_3 = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60$$

eg:- Group of 5 people

Boys = 4 Girls = 7 (at least 3 Girls)

$$\text{No. of groups} = {}^7C_3 {}^4C_2 + {}^7C_4 {}^4C_1 + {}^7C_5 {}^4C_0$$

$$= \frac{7!}{3!(4!)} \times \frac{4!}{2!2!} + \frac{7!}{3!4!} \times \frac{4!}{1!3!} + \frac{7!}{5!2!} \times \frac{4!}{0!4!} = 371$$

Algebraic Structure

$\Rightarrow$ set A (non empty) is called the algebraic structure

$$\text{if } (a * b) \in A$$

$\uparrow$
closure on A (closure property)
(binary operation)

eg:- $(N, +)$

$\uparrow$
Natural
Number

is closed with respect to
'+'
✓

as $1 + 2 = 3 \in N$

$3 + 8 = 11 \in N$

It follows the closure property

$\Rightarrow (N, +)$... algebraic structure.

(Natural
 N)

(Integer)
 Z

(Real)
 R

(Matrix)
 M

$\oplus$ ✓

$\oplus$ ✓

$\oplus$ ✓

$\oplus$ ✓

$\ominus$ ✗

$\ominus$ ✓

$\ominus$ ✓

$\otimes$ ✓

$\oslash$ ✗

$\oslash$ ✗

$\oslash$ ✗

$\otimes$ ✓

$\otimes$ ✓

$\otimes$ ✓

◦ Semi Group

① closure property + ② Associative property

w.r.t '*'

$$AP: (1 * 2) * 3 = 1 * (2 * 3)$$

eg: $(\mathbb{Z}, -)$ holds the closure property but fails to hold AP.

$$\begin{aligned} \Rightarrow (3 - 4) - 5 &= 3 - (4 - 5) \\ \downarrow \quad \quad \quad \downarrow \\ -1 - 5 &= 3 - (-1) \\ -6 &\neq 4 \end{aligned}$$

$$\Rightarrow (\mathbb{N}, +) (\mathbb{N}, \times) (\mathbb{Z}, +) (\mathbb{Z}, \times) (\mathbb{R}, +) \\ (\mathbb{R}, \times) (\mathbb{M}, +) (\mathbb{M}, \times)$$

◦ Monoid

① CP + ② AP + ③ IP

w.r.t '*'

Identity Property:

Identity element $\begin{cases} 0 \text{ in } '+' \\ 1 \text{ in } '\times' \end{cases}$

$$1 + 0 = 1$$

$$5 \times 1 = 5$$

$$\Rightarrow (N, +) : 1, 2, 3 \dots \infty$$

'0' is not there

$$\underline{x + 0 = x}$$

So doesn't hold IP, so not Monoid

$$\Rightarrow (N, \times)$$

$$a \times \underset{\uparrow}{1} = a$$

Its present is 'N'

So, holds on IP, so Monoid.

$\Rightarrow (N, /)$ don't check as it is not Algebraic structure.

$$\Rightarrow (Z, +) (Z, \times) (R, +) (R, \times) (M, +) (M, \times)$$

• Group

	①	②	③	④
$\hookrightarrow$	CP	+ AP	+ IP	+ InP
	w.r.t. to *			

• Inverse Property

$x * x^{-1} = e$ Identity element

set element $\nearrow$

eg: $5 + (-5) = 0$ $5 \times (5^{-1}) = 1$

eg:- $(N, \times)$ It is $[CP + AP + IP]$

but fails to hold InP

as $5 \times 5^{-1} = 1$

$\searrow$ $\frac{1}{5}$ is not in 'N'

So it's not in Group

$\Rightarrow (Z, +)$ It holds $[CP + AP + IP + InP]$

$\Rightarrow$

$$\begin{array}{ccccccc} & 5 & + & (-5) & = & 0 & \\ \uparrow & \uparrow & & \uparrow & & \uparrow & \\ & & * & & & e & \end{array}$$

It is in Group.

$\Rightarrow (R, +) (M, +)$

• Abelian $\rightarrow$ $\begin{array}{ccccc} \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{4} & \textcircled{5} \\ CP & + & AP & + & IP & + & InP & + & CoP \end{array}$
Group $\underbrace{\hspace{10em}}$
 $w.r.t \text{ to } '*'$

Commutative property: $x * y = y * x$

$\Rightarrow (Z, +) (R, +) (M, +)$

◦ Addition Modulo: $+_m$ or $\oplus_m$

$$\text{if } (a + b < m) \quad a +_m b = a + b$$

$$\text{if } (a + b \geq m) \quad a +_m b = (a + b) \% m$$

$$\text{eg: } \{0, 1, 2\}, +_3$$

$$0 +_3 1 = [1 < 3] \text{ so } \rightarrow 1$$

$$1 +_3 2 = [3 = 3] \text{ so } \rightarrow 3 \% 3 = 0$$

$$2 +_3 0 = [2 < 3] \text{ so } \rightarrow 2$$

◦ Multiplication modulo: $*_m$ or $\otimes_m$

$$\text{if } (a * b < m) \quad a *_m b = a * b$$

$$\text{if } (a * b \geq m) \quad a *_m b = (a * b) \% m$$

$$\text{eg: } \{1, 2, 3\} *_4$$

$$1 *_4 2 = [2 < 4] = 2$$

$$2 *_4 3 = [6 > 4] = 6 \% 4 = 2$$

$$3 *_4 1 = [3 < 4] = 3$$

Order of Element

eg: $\{0, 1, 2\} +_3 \quad O(0) = 1$

$$1^1 = 1$$

$$1^2 = 1 +_3 1 = 2 \quad [2 < 3]$$

$$1^3 = 1 +_3 1 +_3 1 = 0 \quad [3 = 3, 0]$$

$$1^4 = 1 +_3 1 +_3 1 +_3 1 = 1$$

$$O(1) = 3$$

$$2^1 = 2$$

$$2^2 = 2 +_3 2 = 1$$

$$2^3 = 2 +_3 2 +_3 2 = 0 \quad [6 - (3+3) = 0]$$

$$2^4 = 2 +_3 2 +_3 2 +_3 2 = 2$$

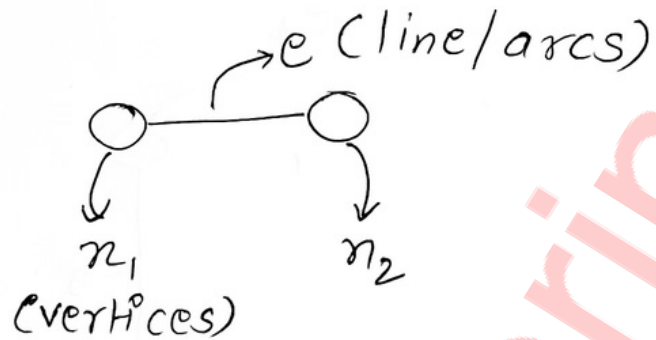
$$O(2) = 3$$

$$\Rightarrow a^{n^{\leftarrow \text{order (small)}}} = e$$

Graph

→ NLDS

└ Nodes
└ Edge



① Directed Graph :



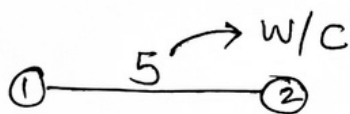
$1 \rightarrow 2$ (1,2)

② Undirected Graph



(1,2) (2,1)

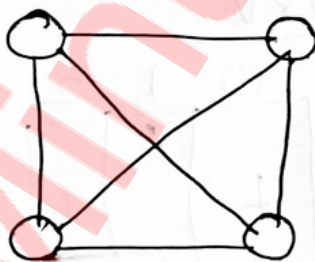
③ weighted Graph



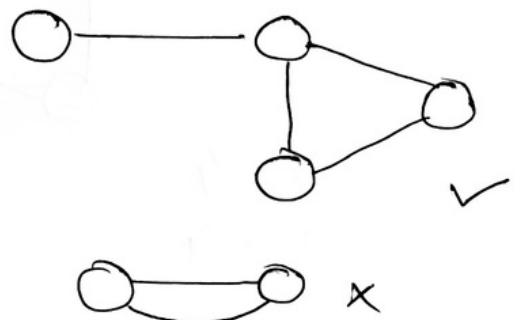
④ Unweighted Graph



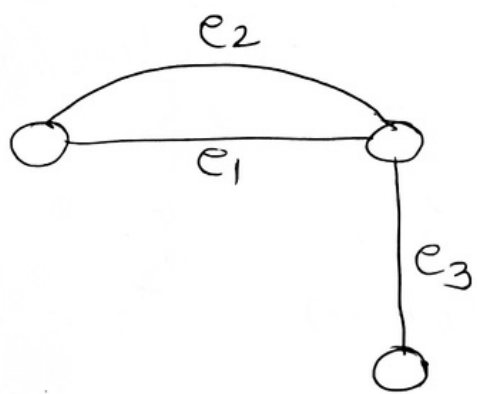
⑤ Complete Graph



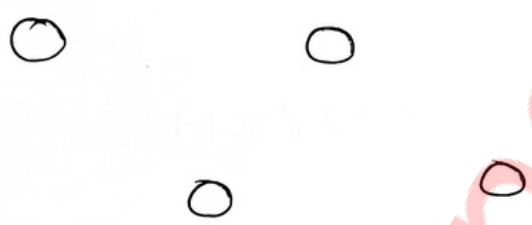
⑥ simple Graph



⑥ Multi Graph



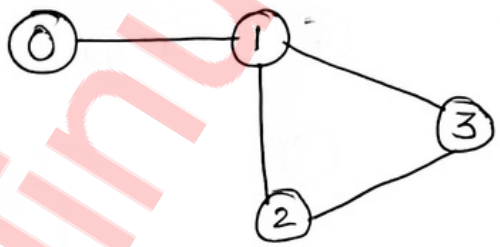
⑦ NULL Graph



⇒ Representation of graph:

- Adjacency Matrix
- Adjacency List

Eg:



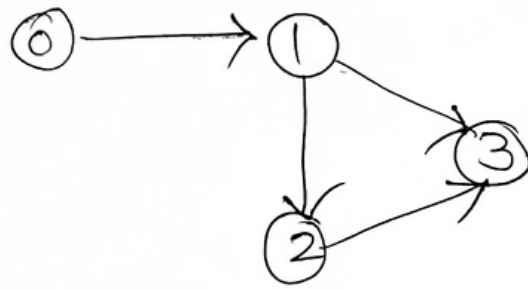
$V = 4$
 $E = 4$

$1 \rightarrow \checkmark$
 $0 \rightarrow \times$

VxV

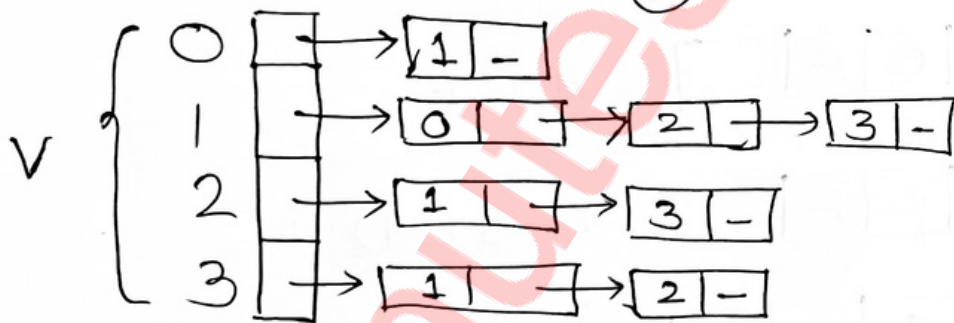
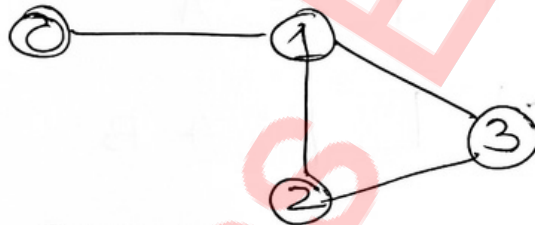
	0	1	2	3
0		1		
1	1		1	1
2		1		1
3		1	1	

Eg

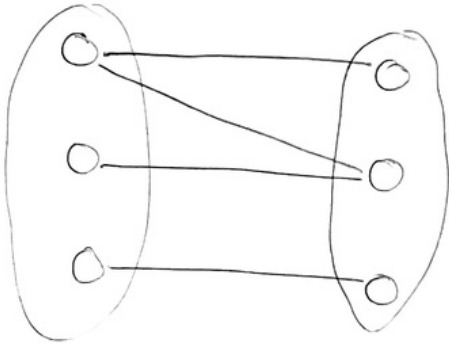


	0	1	2	3
0		1		
1			1	1
2				1
3				

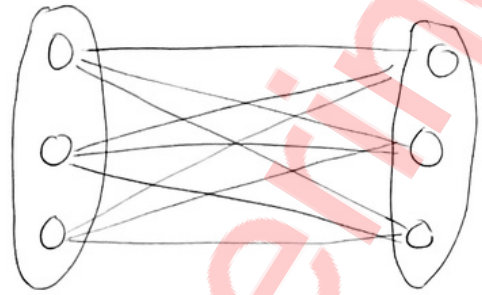
→ Adjacency List



• (Bipartite Graph)

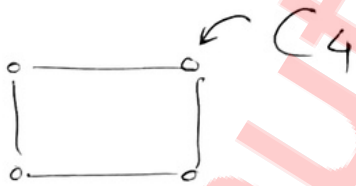
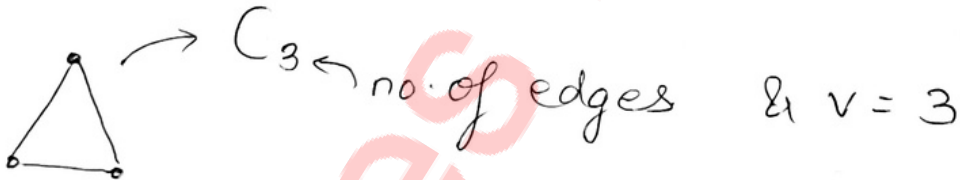


(Complete BpG)

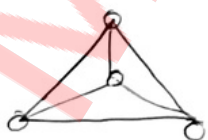


- 2 disjoint sets (partition)
- Same set 'v's not adj.
- Every edge connects 'v' of one set to 'v' of other set.

• Cycle Graph (closed chain)



• Wheel Graph

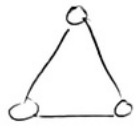


W_4 (center ① surrounded by ③)
 $\downarrow$
 $(n-1)$

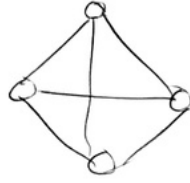
$W_5 \rightarrow$ Center ① surrounded by ④)

• Regular Graph : Degree of all 'V's' is same.

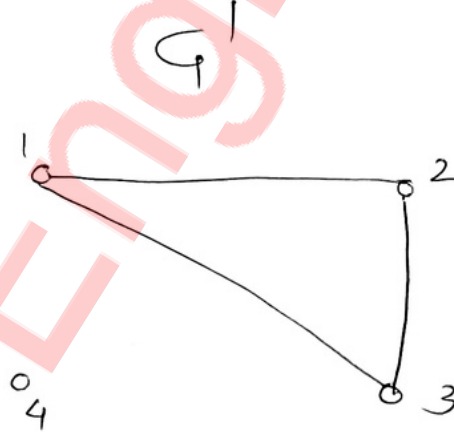
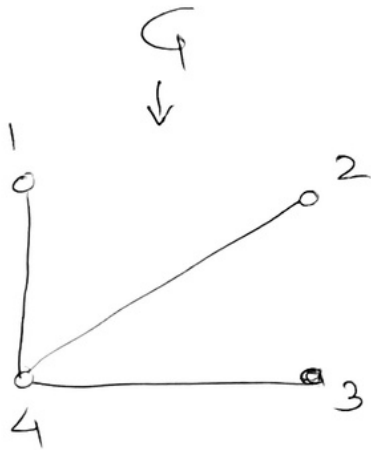
$\Rightarrow$ 2-RG



$\Rightarrow$ 3-RG



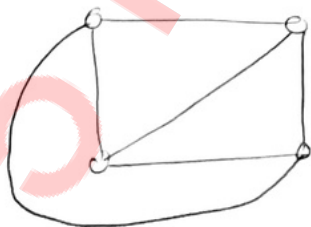
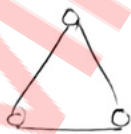
• Complement of a Graph.



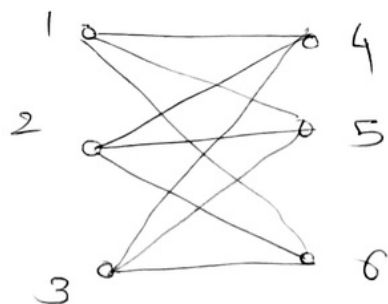
$G \cap G' = \text{only vertex}$

$G \cup G' = \text{Complete Graph}$

• Planar Graph



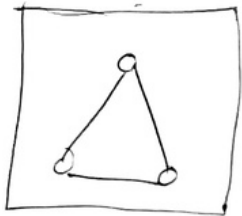
• Non Planar Graph



◦ Euler's Formula:

(faces/region) $r = e - v + 2$

⇒ Consider planar graph only



$$e = 3$$

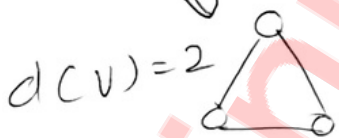
$$v = 3$$

$$\therefore r = 3 - 3 + 2 = 2$$

◦ Euler Graph.

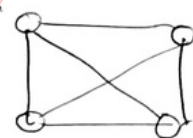
→ start from v_i → go on covering/visiting of edges → & come back to v_i .
(Exactly once)

→ Connected Graph + all v 's have even degree.



$$d(v) = 2$$

Euler Graph

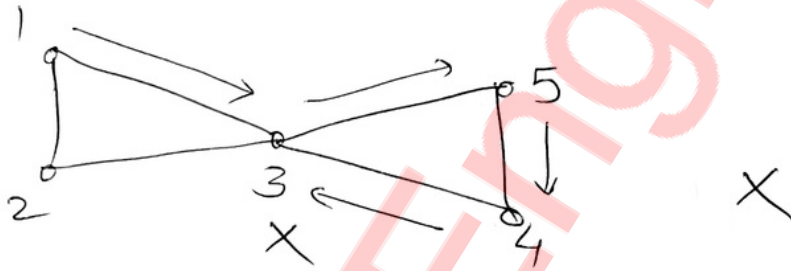
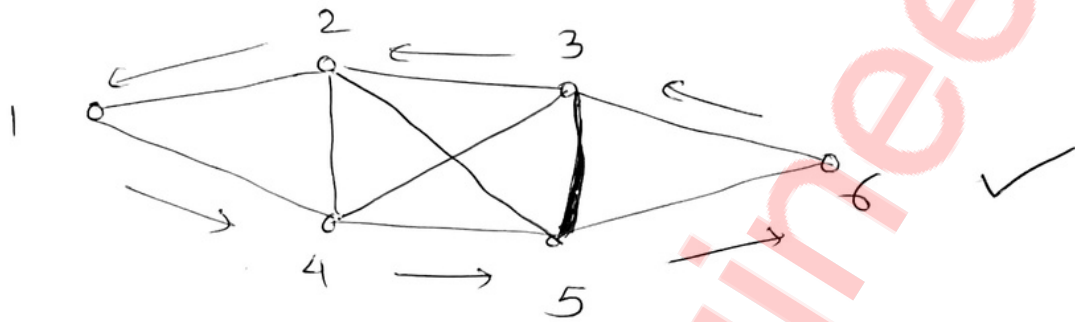


$$d(v) = 3$$

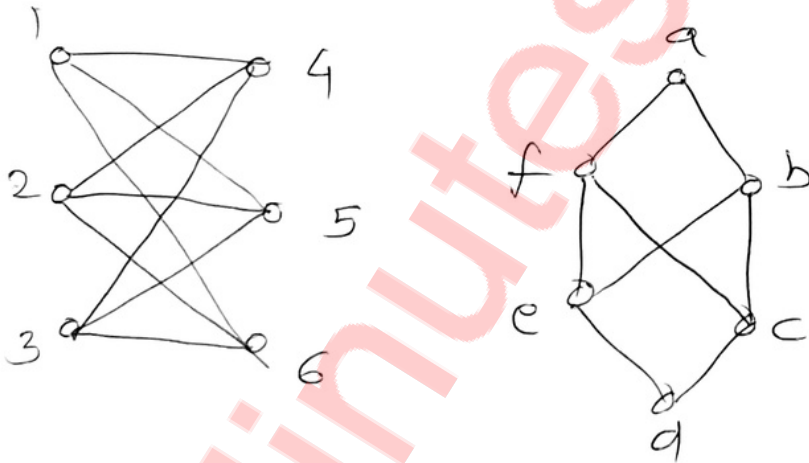
Not a Euler Graph

◦ Hamiltonian Graph

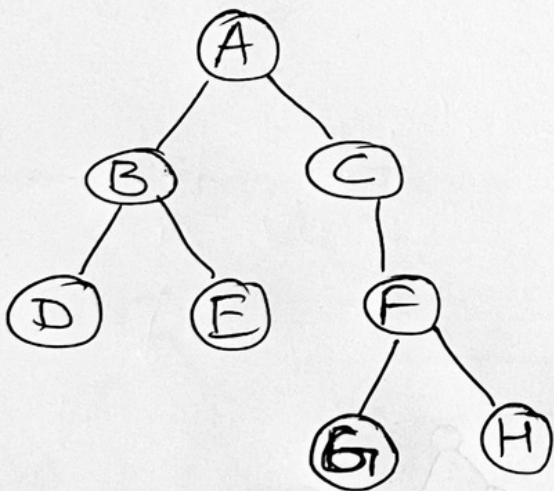
→ start from a V_i and go on visiting all ' V ' exactly once & come back to V_i



◦ Isomorphism



→ Tree



• root = A

• leaf = D, E, G, H

• Parent = A, B, C, F

• children = B, C, D, E, F, G, H

• Non-LN = $\overline{LN}$

• Path = A → C → F → H
(A → H)

• Ancestor & Decendant
(Before node) (After node)
(A) (F, H)

• Sibling → D, E ✓
B, C ✓
E, F X

• Depth: Root → node

(A) = 0 (D) = 2

(B) = 1 (E) = 2

(G) = 3 (H) = 3

• Height: node → leaf*

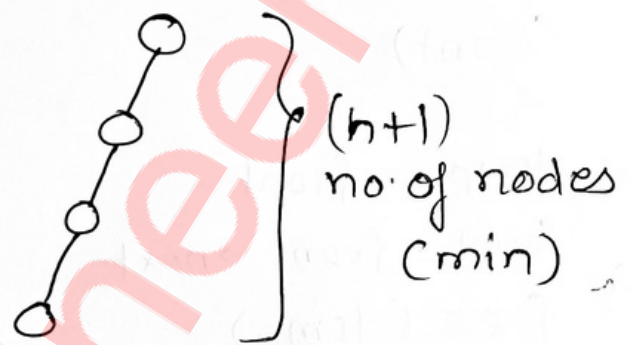
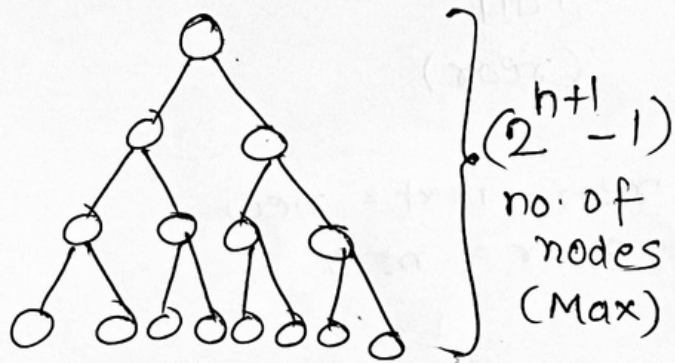
(A) = 3 (C) = 2

(F) = 1 (E) = 0

• Degree of node → A = 2
B = 2
D = 0

→ Binary Tree

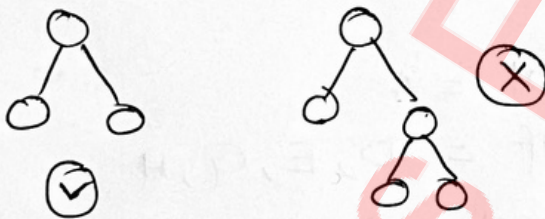
↳ can have at most 2 children



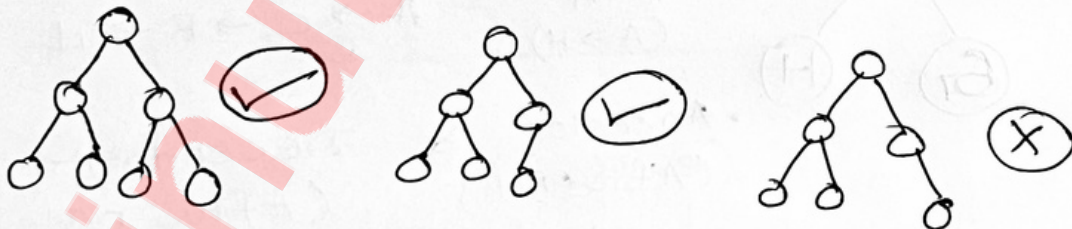
for $h=3$, $2^{3+1} - 1 = 15$

for $h=3$, $3+1 = 4$

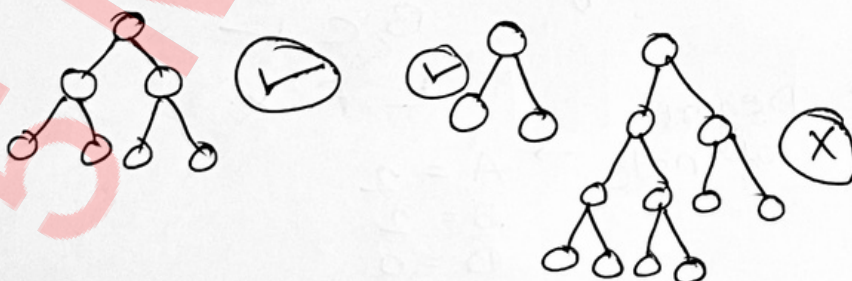
① Full BT (0 or 2)



② Complete BT

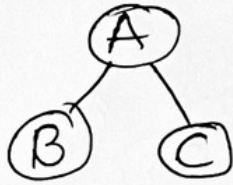


③ Perfect BT



⇒ Tree Traversal

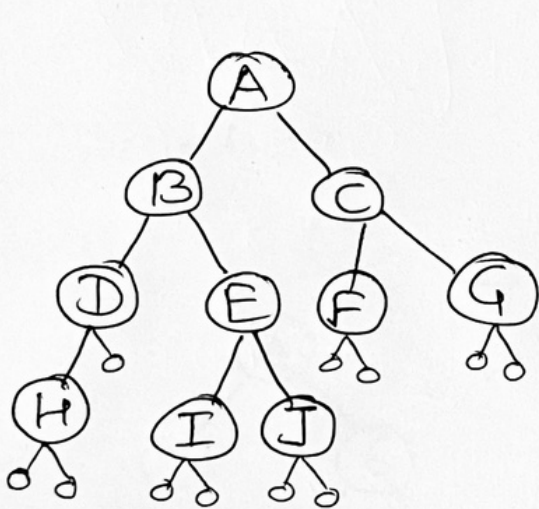
- Inorder ($l \rightarrow R \rightarrow r$)
- Preorder ($R \rightarrow l \rightarrow r$)
- Postorder ($l \rightarrow r \rightarrow R$)



Pre
ABC

In
BAC

Post
BCA

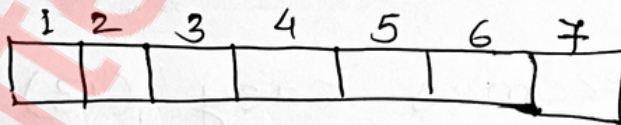
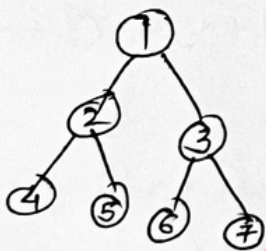


①
In

②
Pre

③
Post

→ Binary Tree (Using Arrays)



Root at $\rightarrow 1$

Left child $\rightarrow 2 * i$

Right child $\rightarrow 2 * i + 1$

for eg:- $i = 3$ $\rightarrow L = 2 \times 3 = 6$
 $\rightarrow R = 2 \times 3 + 1 = 7$

° Prefix Code : No sequence in the set of sequences is a prefix of other sequence.

eg: {001, 000, 110, 01}

eg: {001, 0010, 010}

° Huffman Algorithm

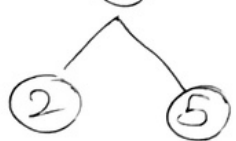
eg:- 10, 8, 12, 15, 5, 2

① Ascending order

② ⑤ ⑧ ⑩ ⑫ ⑮

② Combine first 2

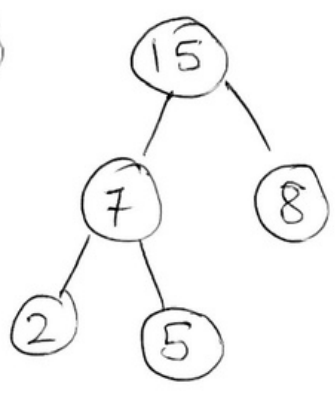
⑦ → place it in the above order



→



③



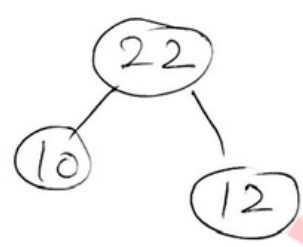
⑩

⑫

⑮

⑯

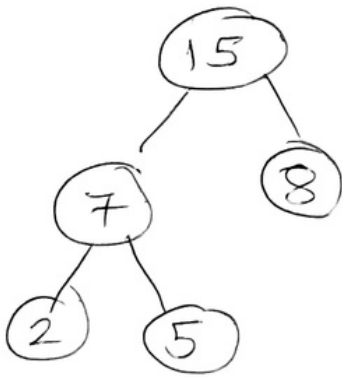
④



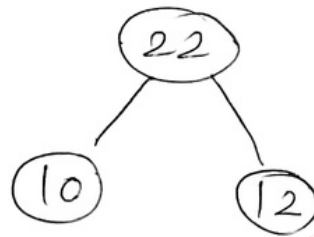
⑮

⑯

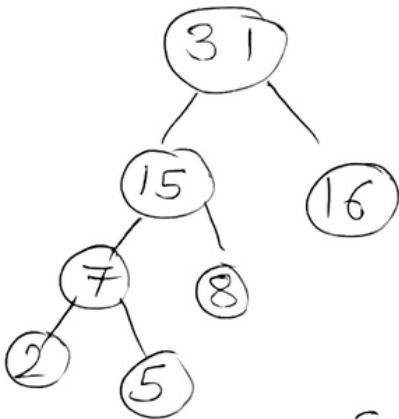
②②



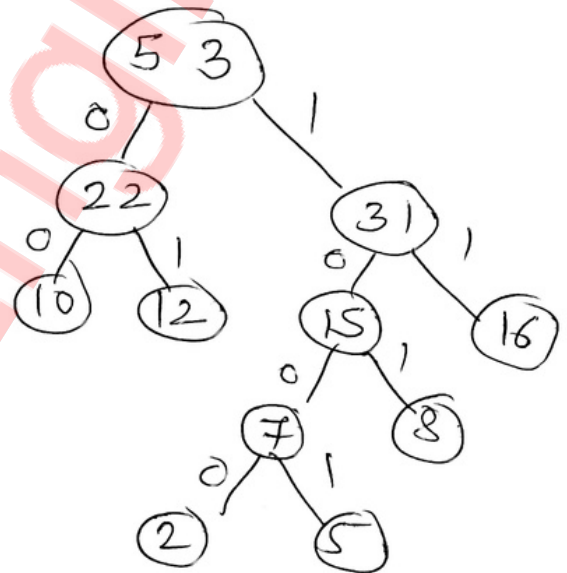
16



5



so finally.



$$10 \Rightarrow 00$$

$$12 \Rightarrow 01$$

$$2 \Rightarrow 1000$$

$$5 \Rightarrow 1001$$

$$8 \Rightarrow 101$$

$$16 \Rightarrow 11$$

$$\text{weight} = 10 \times 2 + 12 \times 2 + 2 \times 4 + 5 \times 4 + 8 \times 3 + 16 \times 2$$

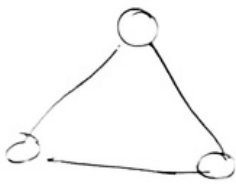
$$= 20 + 24 + 8 + 20 + 24 + 32$$

$$= 128$$

Spanning Tree

- ① It covers all 'v's'
- ② No cycle
- ③ Has total $(n-1)$ edges → no. of 'v's'.

Graph ↘

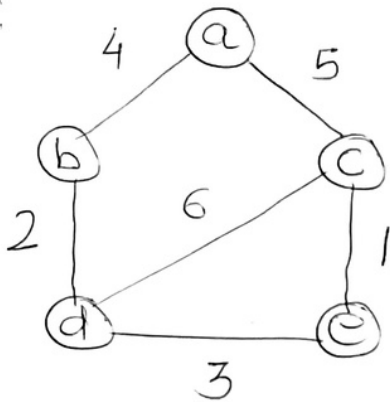


STs ↘



• Prim's algorithm & Kruskal algorithm

eg:



eg:

